

PhoenixHRMS

SUMMARY

The Learning Management Module provides a comprehensive enterprise-grade learning and development ecosystem that enables organizations to build workforce capabilities, support career growth, ensure compliance, and drive continuous improvement.

DOCUMENT METADATA

Version: 1.0 Status: Draft Module ID: LMS Document Type: Functional Module Specification Related Documents: - 15-employee-management-module.md - 20-performance-management-module.md - 11-database-design.md - 14-openapi-spec.md

PURPOSE

The Learning Management Module enables organizations to create, manage, deliver, track, and analyze employee learning and development activities.

The module supports: - Learning Catalogs - Courses - Learning Paths - Certifications - Compliance Training - Instructor-Led Training - Assessments - Skill Development

BUSINESS OBJECTIVES

- Improve workforce capabilities - Reduce training administration effort - Increase compliance readiness - Support career development - Improve employee engagement - Enable continuous learning culture

MODULE SCOPE - INCLUDED

Course Management Learning Catalog Learning Paths Assessments Quizzes Certifications Instructor-Led Training Virtual Training Compliance Training Skill Frameworks Training Requests Learning Analytics AI Learning Recommendations

MODULE SCOPE - EXCLUDED

Payroll Processing Attendance Tracking Performance Reviews
These consume LMS outputs.

LEARNING LIFECYCLE

Course Created → Published → Assigned → Enrollment → Learning → Assessment → Certification → Skill Update → Performance Impact

LEARNING CATALOG

Purpose: Central repository of learning resources.

Categories: Technical Skills Leadership Compliance Product Training Soft Skills Sales Training Safety Training Custom Categories

Features: Browse Search Filter Recommendations Bookmarks Ratings Reviews

COURSE MANAGEMENT

Course Types: Self-Paced Instructor-Led Virtual Classroom Blended Learning Microlearning Certification Course Assessment Only

Course Attributes: Title Description Category Difficulty Level Duration Language Tags Thumbnail Status

Statuses: Draft Published Archived Retired

LEARNING CONTENT

Supported Formats: PDF Video Audio PPT SCORM xAPI HTML External Links Documents

Features: Versioning Preview Download Control Access Restrictions

LEARNING PATHS

Purpose: Structured learning journeys.

Example: Junior Developer → Programming Basics → Git Fundamentals → PHP Fundamentals → Laravel Framework → Project Assignment

Features: Mandatory Sequence Optional Courses Completion Rules Progress Tracking

ENROLLMENT MANAGEMENT

Enrollment Types: Self Enrollment Manager Enrollment HR Enrollment Bulk Enrollment Auto Enrollment

Enrollment Statuses: Assigned Enrolled In Progress Completed Expired Cancelled

COURSE ASSIGNMENTS

Assignment Criteria: Department Designation Location Grade Role Skill Gap Performance Plan

Assignment Types: Mandatory Recommended Optional

INSTRUCTOR MANAGEMENT

Instructor Types: Internal Trainer External Trainer Guest Trainer Vendor Trainer

Instructor Profile: Name Skills Certifications Experience Ratings Availability

CLASSROOM TRAINING

Features: Session Scheduling Capacity Management Attendance Tracking Resource Allocation Waitlists Feedback Collection

Session Statuses: Scheduled In Progress Completed Cancelled

VIRTUAL TRAINING

Features: Video Integration Live Sessions Recordings Attendance Tracking Interactive Polls Q&A

Integrations: Zoom Microsoft Teams Google Meet Webex

ASSESSMENTS

Assessment Types: Quiz Assignment Practical Evaluation Case Study Project Submission Certification Exam

Features: Question Banks Randomization Time Limits Passing Scores Retakes

QUIZ MANAGEMENT

Question Types: Multiple Choice Multiple Select True/False Short Answer Essay Matching Fill In The Blank

Features: Scoring Negative Marking Difficulty Levels Question Pools

CERTIFICATION MANAGEMENT

Certification Attributes: Name Issuer Issue Date Expiry Date Validity Period Certification Number

Features: Renewals Expiry Notifications Verification Compliance Tracking

COMPLIANCE TRAINING

Purpose: Meet regulatory requirements.

Examples: Workplace Safety Data Privacy Code Of Conduct Anti-Harassment Cyber Security Industry Compliance

Features: Mandatory Assignments Deadlines Escalations Compliance Reporting

SKILL FRAMEWORK

Skill Categories: Technical Leadership Communication Management Functional Domain Behavioral

Skill Levels: Beginner Intermediate Advanced Expert

SKILL ASSESSMENTS

Inputs: Course Completion Assessment Results Manager Evaluation Certification Status

Outputs: Skill Scores Skill Gaps Development Recommendations

LEARNING RECOMMENDATIONS

Recommendation Sources: Role Department Career Path Performance Review Skill Gap Analysis AI Suggestions

Examples: Leadership Basics Recommended For New Managers

TRAINING REQUESTS

Workflow: Employee → Manager → HR → Approved → Enrollment

Request Types: External Training Conference Workshop Certification Custom Learning

LEARNING ANALYTICS

Metrics: Enrollment Rate Completion Rate Course Effectiveness Assessment Scores Certification Rate Skill Growth

Learning Hours Compliance Rate

LEARNING REPORTS

Course Completion Report Certification Report Compliance Report Skill Gap Report Learning Path Progress

Instructor Effectiveness Assessment Results Learning Hours Report

Formats: PDF Excel CSV

SOCIAL LEARNING

Features: Discussion Forums Course Reviews Peer Recommendations Knowledge Sharing Learning Communities

Future Enhancement: Enterprise Learning Network

GAMIFICATION

Features: Points Badges Achievements Leaderboards Learning Challenges Milestones

Objectives: Increase engagement Improve completion rates

AI LEARNING FEATURES

Features: Course Recommendations Learning Path Generation Skill Gap Analysis Content Summarization Learning Assistant Career Development Suggestions

Governance: Recommendations only. Human control retained.

LEARNING NOTIFICATIONS

Course Assigned Course Due Soon Course Completed Certification Expiring Assessment Scheduled Assessment Passed Assessment Failed Learning Path Completed

LMS APIS

Courses: GET /courses POST /courses PATCH /courses/{id}

Learning Paths: GET /learning-paths POST /learning-paths

Enrollments: GET /enrollments POST /enrollments

Assessments: GET /assessments POST /assessments

Certifications: GET /certifications POST /certifications

DATABASE TABLES

Catalog Domain: courses course_categories course_content

Learning Path Domain: learning_paths learning_path_courses

Enrollment Domain: enrollments enrollment_progress

Assessment Domain: assessments questions assessment_attempts assessment_results

Certification Domain: certifications employee_certifications

Skills Domain: skills employee_skills skill_assessments

Training Domain: training_sessions session_attendees instructors

PERMISSIONS MATRIX

lms.view lms.manage_courses lms.manage_learning_paths lms.manage_assessments lms.manage_certifications
lms.manage_enrollments lms.manage_instructors lms.analytics.view lms.export

SECURITY REQUIREMENTS

Content Access Control Certification Integrity Assessment Security Tenant Isolation RBAC Enforcement Audit Logging

AUDIT REQUIREMENTS

Course Created Course Published Enrollment Assigned Assessment Submitted Certification Issued Learning Path Completed Training Session Scheduled

Retention: 7 Years Minimum

INTEGRATIONS

Employee Module Performance Module Recruitment Module Notification Module Video Conferencing Platforms
Content Providers SCORM Providers AI Module

PERFORMANCE INTEGRATION

Outputs: Skill Growth Course Completion Certification Status Development Progress

Consumed By: Performance Reviews Career Planning Succession Planning

DASHBOARD WIDGETS

Assigned Courses Courses Due Soon Learning Hours Certifications Expiring Skill Progress Compliance Training
Status Top Learners Course Completion Rate

PERFORMANCE REQUIREMENTS

Course Search: < 1 Second Assessment Submission: < 2 Seconds Analytics Dashboard: < 3 Seconds

FUTURE ENHANCEMENTS

Version 2 AI Tutor Adaptive Learning Paths Skills Marketplace Learning Experience Platform (LXP) Knowledge
Graph Career Navigator Learning Copilot Virtual Reality Training Augmented Reality Training

SUCCESS METRICS

Course Completion Rate: > 85% Compliance Completion Rate: > 95% Certification Renewal Rate: > 90% Learning
Engagement Rate: > 80%

CONCLUSION

The module integrates learning content, assessments, certifications, skills, development plans, and AI-powered recommendations into a unified learning platform while maintaining scalability, security, auditability, and tenant isolation.